

UNIT 2

AI Techniques for Environmental Modelling

Learning Objectives

After studying this unit, students will be able to:

- Explain regression techniques used in environmental prediction.
 - Understand classification techniques such as Decision Trees and k-Nearest Neighbour (k-NN).
 - Explain clustering using K-Means.
 - Understand the concept of Artificial Neural Networks (ANN).
 - Describe model training, validation, and testing.
 - Evaluate AI models using performance indicators.
 - Identify limitations and uncertainties in AI models.
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2.1 Regression Techniques for Prediction

(Linear Regression Concept)

What is Regression?

Regression is a **supervised machine learning technique** used to predict a **continuous numerical value** based on one or more input variables.

Unlike classification, regression predicts values such as:

- Temperature
- Air Quality Index (AQI)
- Noise Level (dB)
- River pH
- Dissolved Oxygen
- Rainfall
- Water Demand

Definition

Regression is a machine learning technique that predicts continuous values by finding the relationship between input variables and output variables.

Environmental Engineering Examples

Input	Output Prediction
Traffic + Weather	Noise Level
PM2.5 + Humidity	AQI
Rainfall	River Water Level
Temperature	Ozone Concentration
Population	Solid Waste Generation

Types of Regression

Although many regression methods exist, diploma students need to understand the basic concept of **Linear Regression**.

Examples of regression methods include:

- Linear Regression
 - Multiple Linear Regression
 - Polynomial Regression
 - Ridge Regression
 - Lasso Regression
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Linear Regression

Linear Regression assumes a **linear relationship** between the input variable (X) and the output variable (Y).

The relationship is represented by a straight line.

Mathematical Equation

$$Y = mX + c$$

Where:

- **Y** = Predicted value
- **X** = Input variable
- **m** = Slope of the line
- **c** = Intercept

Example

Predict Noise Level using Traffic Volume.

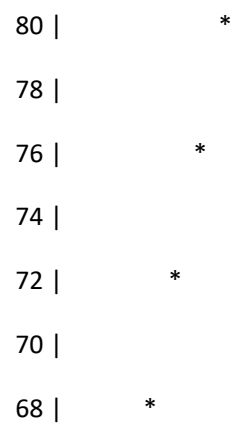
Traffic Volume (Vehicles/hr) Noise (dB)

100	62
200	67
300	72
400	77

Linear regression identifies the best-fit line and predicts the noise level for other traffic volumes.

Graph

Noise Level



66 |

64 | *

62 | *

100 200 300 400 Traffic Volume

Advantages

- Simple to understand
- Fast computation
- Easy interpretation
- Suitable for prediction problems

Limitations

- Works well only for linear relationships
- Sensitive to outliers
- Cannot capture complex patterns

Environmental Applications

- AQI prediction
- Noise prediction
- Water demand estimation
- Rainfall estimation
- Energy consumption forecasting

2.2 Classification Techniques

Classification is another important supervised machine learning technique.

Instead of predicting numbers, it predicts **categories or classes**.

Definition

Classification assigns data into predefined groups based on its characteristics.

Examples

Input	Output
Waste Image	Plastic
Water Sample	Safe/Unsafe
Air Quality	Good/Poor
Tree Image	Healthy/Diseased
Noise Source	Traffic/Industrial

Difference Between Regression and Classification

Regression	Classification
Predicts numbers	Predicts categories
Example: AQI = 115	Example: AQI = Moderate
Continuous output	Discrete output

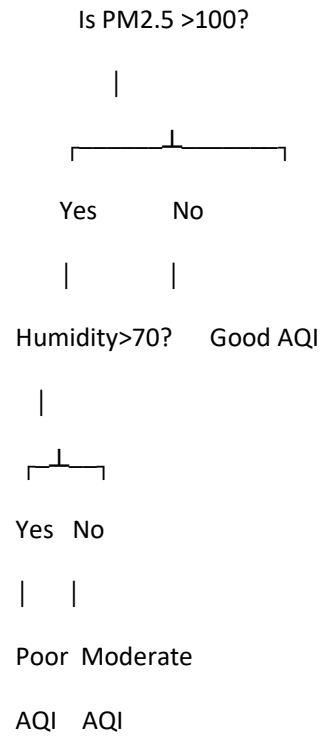
Decision Tree (Basic Concept)

What is a Decision Tree?

A Decision Tree is a flowchart-like machine learning model that makes decisions by asking a series of questions.

Each question divides the data into smaller groups until a final decision is reached.

Structure



Advantages

- Easy to understand
 - Simple visualization
 - Handles numerical and categorical data
 - Fast prediction
-

Limitations

- Can become very large
 - Sensitive to noisy data
 - May overfit
-

Environmental Applications

- Water quality classification
- Waste segregation

- Pollution source identification
 - Land-use classification
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k-Nearest Neighbour (k-NN)

Definition

k-NN classifies a new data point based on the majority class of its nearest neighbours.

The value of **k** represents the number of neighbours considered.

Example

Suppose we want to classify a water sample.

Nearby samples:

Safe

Safe

Unsafe

Safe

If $k = 3$

Majority = Safe

Therefore,

New sample → Safe

Working Steps

1. Select k .
 2. Calculate distance from all data points.
 3. Identify nearest neighbours.
 4. Majority class becomes prediction.
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Advantages

- Very simple
 - No training required
 - Good for small datasets
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Limitations

- Slow for large datasets
 - Sensitive to irrelevant features
 - Performance depends on k value
-

Environmental Applications

- Water quality classification
 - Soil classification
 - Pollution zone identification
 - Waste type identification
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Comparison of Classification Techniques

Feature	Decision Tree k-NN	
Training Required	Yes	Minimal
Easy to Interpret	Excellent	Moderate
Prediction Speed	Fast	Slow
Works with Large Data	Yes	Limited

2.3 Clustering for Environmental Data Grouping

(K-Means Concept)

What is Clustering?

Clustering is an **unsupervised machine learning technique**.

It groups similar data without knowing the correct answers beforehand.

Definition

Clustering automatically groups similar observations into clusters based on similarity.

Environmental Examples

- Cities with similar pollution
 - Rivers having similar water quality
 - Areas with similar rainfall
 - Noise hotspots
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K-Means Clustering

K-Means is the most popular clustering algorithm.

"K" represents the number of clusters.

Working Steps

Step 1

Select number of clusters (K).

↓

Step 2

Randomly place cluster centres.

↓

Step 3

Assign each data point to nearest centre.

↓

Step 4

Calculate new cluster centres.



Step 5

Repeat until centres stop changing.

Example

Suppose pollution data from ten cities.

AI groups them into:

Cluster 1 → Highly Polluted

Cluster 2 → Moderately Polluted

Cluster 3 → Clean Cities

Environmental Applications

- Air pollution zoning
 - Water quality grouping
 - Noise hotspot identification
 - Industrial pollution classification
 - Land-use analysis
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Advantages

- Simple
 - Fast
 - Good visualization
 - Useful for pattern discovery
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Limitations

- Number of clusters must be chosen beforehand

- Sensitive to outliers
- Results depend on initial cluster centres

2.4 Introduction to Artificial Neural Networks (ANN)

What is an Artificial Neural Network?

Artificial Neural Network (ANN) is a machine learning model inspired by the human brain.

It consists of interconnected processing units called neurons.

ANN is capable of learning complex relationships from data.

Structure

Input Layer

↓

Hidden Layer

↓

Hidden Layer

↓

Output Layer

Example

Input:

- Temperature
- Humidity
- Wind Speed
- PM2.5

↓

ANN

↓

Output:

Tomorrow's AQI

Advantages

- High prediction accuracy
 - Learns complex relationships
 - Handles nonlinear problems
 - Excellent for environmental prediction
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Limitations

- Requires large datasets
 - High computational cost
 - Difficult to interpret ("black-box" model)
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Environmental Applications

- AQI forecasting
- Flood prediction
- Noise modelling
- Climate forecasting
- Water quality prediction

2.5 Model Training, Validation, and Testing

Why Divide Data?

To ensure that the AI model performs well on new, unseen data.

Training Data

Used to teach the model.

Usually **70–80%** of the dataset.

Validation Data

Used to tune model parameters and avoid overfitting.

Usually **10–15%** of the dataset.

Testing Data

Used only after training to evaluate final performance.

Usually **10–20%** of the dataset.

Workflow

Collected Data



Training Set



Validation Set



Testing Set



Final Model

Example

1000 environmental records

Training → 700

Validation → 150

Testing → 150

Overfitting

The model memorizes training data but performs poorly on new data.

Underfitting

The model is too simple and cannot learn patterns.

Good Model

Performs well on both training and testing datasets.

2.6 Performance Indicators

Why Evaluate Models?

Evaluation tells us how accurate and reliable the AI model is.

Accuracy

Accuracy measures the percentage of correct predictions.

Formula

Accuracy =

Correct Predictions ÷ Total Predictions × 100

Example

100 samples

Correct predictions = 92

Accuracy

=92%

Mean Square Error (MSE)

Regression models cannot use accuracy directly.

Instead they use error measurements.

Definition

Mean Square Error measures the average squared difference between actual and predicted values.

Formula

[
MSE = $\frac{\sum (\text{Actual} - \text{Predicted})^2}{n}$
]

Example

Actual AQI =100

Predicted AQI =95

Error =5

Squared Error =25

Smaller MSE means a better model.

Comparison

Indicator Used For

Accuracy Classification

MSE Regression

2.7 Limitations and Uncertainties in AI Models

Why Do AI Models Make Errors?

AI models depend on data and assumptions. Poor data or changing environmental conditions can reduce accuracy.

Major Limitations

1. Poor Data Quality

Missing or incorrect measurements lead to inaccurate predictions.

2. Limited Training Data

Small datasets may not represent real-world conditions.

3. Overfitting

Model performs well on training data but poorly on new data.

4. Underfitting

Model fails to capture important relationships.

5. Sensor Errors

Faulty environmental sensors produce unreliable inputs.

6. Climate Variability

Environmental systems change over time, making predictions more difficult.

7. Black-Box Nature

Complex AI models such as ANN often do not explain how they arrive at a decision.

Sources of Uncertainty

- Instrument errors
 - Human errors in data collection
 - Missing values
 - Seasonal variations
 - Climate change
 - Unexpected events (e.g., industrial accidents, wildfires)
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Reducing Uncertainty

- Use high-quality data
- Collect sufficient observations
- Perform data cleaning
- Validate models properly
- Regularly update models with new data
- Use multiple evaluation metrics

Summary of Unit 2

Topic	Key Concept
Regression	Predicts continuous values (e.g., AQI, noise level)
Linear Regression	Uses a straight-line relationship between variables
Classification	Assigns data into predefined categories
Decision Tree	Makes decisions using a flowchart of rules
k-NN	Classifies based on the nearest neighbours
Clustering	Groups similar data without labels
K-Means	Popular clustering algorithm using K clusters
Artificial Neural Network	Brain-inspired model for complex prediction
Training	Teaches the model using historical data
Validation	Tunes the model and checks performance during development
Testing	Evaluates the final model on unseen data
Accuracy	Performance metric for classification
Mean Square Error (MSE)	Performance metric for regression
Limitations	AI performance depends on data quality, model selection, and environmental uncertainty

This material is suitable for approximately **35–45 pages** when formatted with diagrams, illustrations, solved examples, practice questions, and classroom notes. It aligns well with a **Diploma Environmental Engineering** curriculum and provides sufficient conceptual depth without requiring advanced mathematics.